

CCST9012 Tutorial Week 2

Two Sphere Universe

In Lecture 3, we discussed the round Earth hypothesis with a focus on the daily path of the sun on the Solstices and Equinoxes. In Lecture 4, we tracked the path of the Sun through the 12 zodiac constellations along a path called the *ecliptic*. The introduction of the ecliptic expands our discussion from Lecture 3 so that we can predict the daily path of the sun on any day of the year. This completes our understanding of the Two Sphere Universe model.

The Two Sphere Universe comprises:

- A round Earth that doesn't move and is in the center of the Universe. A position on Earth can be located by latitude and longitude.
- A celestial sphere which rotates around the Earth once every 23 hours and 56 minutes (sidereal day). The direction of rotation is from East to West. The axis of rotation is defined by the North and South Celestial Poles. A line between these celestial poles, passes through the Earth's North and South Poles. The stars are located on the surface of the celestial sphere and can be located according to their declination and right ascension. The Sun also participates in the rotation of the celestial sphere. The celestial sphere is much larger than Earth, such that the Earth seems to be a single point.
- The ecliptic defines the path of the Sun's annual motion. It is a circle on the celestial sphere. It is inclined at 23.4 degrees to the Celestial Equator. The solstices and equinoxes can be defined as points on the ecliptic corresponding to the location on the celestial sphere where the sun can be seen on those days. The Summer Solstice is the point where the ecliptic gets nearest the North Celestial Pole. The Equinoxes are the intersections of the ecliptic with the Celestial Equator. The sun completes a circle on the ecliptic once a year.

Consider the simplicity of the model. It has only three parameters: the speed of the celestial sphere's rotation, the speed of the sun's annual motion, the angle the ecliptic makes with the celestial equator. Today's interactive simulation will allow us to change the parameters of the Two-Sphere universe in hopes of a deeper understanding of how it works.

1 TWO PERSPECTIVES

Two perspectives are shown. In Perspective 1, changing the latitude slider rotates the horizon, zenith and Nadir around the celestial sphere. It shows the model in the Equatorial System. In Perspective 2, changing the latitude rotates the celestial sphere within the horizon. It shows the model with reference to an observer's horizon, zenith, nadir.

1. **Take a few moments to convince yourself that these are completely equivalent. Which to you prefer and why?**

2 THE EARTH

Check your understanding.

2. In this model, how long does it take the *Earth* to complete a full rotation?

Trick question! In this model, the Earth does not move at all.

3 THE STARS

In either perspective, check the Daily Motion box. This will cause the Celestial Sphere to rotate but not as it does in real time.

3. In real time, how long should it take for the celestial sphere to complete a full rotation?

The celestial sphere completes a full rotation once every *sidereal day* (23 hours, 56 minutes). This is slightly shorter than a *solar day* (24 hours).

4. Show that which stars will rise/set and which are *circumpolar* depends on the latitude of an observer. How did you demonstrate this?

This can be demonstrated by locating a circumpolar star and then shifting the latitude slider further south until that star sets below the horizon.

4 THE SUN

Uncheck the Daily Motion box. Check the Annual Motion box. This will cause the Sun to move along the ecliptic. Again it does not do so in real time.

5. In real time, how long does it take for the Sun to complete a full circle around the ecliptic?

The sun completes a full rotation once every year.

In the simulation, you can speed up or slow down the speed of the annual motion by adjusting the Year Length. Choose a value for Year Length between 50 and 100. This will be fast enough that you can still see the annual motion but slow enough so that the Sun's annual motion will be much slower than its daily motion.

Make sure both Daily Motion and Annual Motion boxes are checked. Observe the Sun's motion.

6. Compare the Sun's motion to the Celestial Equator.

The Sun's motion seems parallel to the celestial equator on a daily basis, however the second motion causes it to shift north and south along the celestial poles.

7. Adjusting only the latitude, create a midnight sun and a zenith passage. (You may turn off the annual motion and change the sun's position on the ecliptic using the Annual Motion slider)

To do this, stop the annual motion when the sun is located near the point on the ecliptic labeled SS for Summer Solstice. This corresponds to 270 degrees on the annual motion slider. For latitudes further north of the arctic circle, you will get a midnight sun. For a latitude equal to the the tropic of cancer you will get a zenith passage.

The sun's annual motion is from West to East while the daily motion of the celestial sphere is from East to West.

8. Why does the sun rise in the east?

The sun rises in the East because the rotation of the celestial sphere is much faster than the motion of the sun along the ecliptic. As the sun participates in both motions, from earth's perspective it is always moving from east to west, even though it moves from west to east through the zodiac constellations.

9. Is it possible to change the speed of the annual motion so that the sun rises in the west?

The annual motion can be sped up by decreasing the Year Length. If it is set to '1', then the speed of the annual motion along the ecliptic is the same as the speed of the celestial sphere's rotation. This makes the sun appear to stay in approximately one place from the Earth's perspective. (The actual shape of the sun's motion is called an analemma). If you could set the Year Length to less than 1, the speed of the Sun's motion along the ecliptic would be faster than the motion of the celestial sphere and it would rise in the West. One of Mars' moons does this.

5 CHANGE THE WORLD

Perspective 1 allows you to change the *obliquity* or the angle that the ecliptic makes with the celestial equator.

Change it to 0 degrees.

10. Compare the length of daylight for different days of the year.

Always the same.

11. Compare the sunrise/sunset locations for different days of the year.

Always the same.

12. Compare the noon altitude for different days of the year.

Always the same.

13. Is there any sign of season that you can detect?

Not in terms of our daily lives. However, the sun would still be seen to visit different constellations throughout a year.

14. Change the obliquity until you can find a location on Earth that has both a zenith passage and a midnight sun.

This can be done for any value of obliquity greater than 45 degrees.

15. Find the value(s) for the obliquity for which all locations have a zenith passage and a midnight sun.

If the angle were 90 degrees, all locations would have both a zenith passage and a midnight sun.

